

Based on the video title "Tom meets Patrick in the parking lot", here's the transcript with speaker names replaced:

Tom: Good morning Patrick, how are you?
 Patrick: Good, good. How are you Tom?
 Tom: Pretty good. Going to work?
 Patrick: Yes. Busy day. How are your kids? Do they go to school?
 Tom: Oh they are too young for that. I sent them to daycare earlier today.
 Patrick: Oh yeah, I forgot about that.

B.2 AUTOFILLING SPEAKER ROLES

In some applications, knowing the roles of each speaker is useful enough. For example, it would be helpful to distinguish the doctor from the patient in medical transcripts [25], and to distinguish the customer from the agent in call center conversations. This task can be handled with LLM usually without additional metadata about the conversation. We show an example below.

Prompt:

Replace the <spk:?> tokens from the below conversation with roles like teacher, student, doctor, patient, etc.

<spk:1> Hi, how can I help you today?
 <spk:2> Hi, I recently often feel quite dizzy at work.
 <spk:1> Do you have any of these symptoms? Like coughing, fever, or running nose?
 <spk:2> No, I don't think so.
 <spk:1> What do you usually eat for breakfast, lunch and dinner?
 <spk:2> I'm on a diet, so not eating very much. Mostly just one slice of toast in the morning, and eating salad for lunch and dinner.

Completion:

Doctor: Hi, how can I help you today?
 Patient: Hi, I recently often feel quite dizzy at work.
 Doctor: Do you have any of these symptoms? Like coughing, fever, or running nose?
 Patient: No, I don't think so.
 Doctor: What do you usually eat for breakfast, lunch, and dinner?
 Patient: I'm on a diet, so not eating very much. Mostly just one slice of toast in the morning, and eating salad for lunch and dinner.

B.3 REPLACING THE ORCHESTRATION MODULE

In Fig. 1, we have explained how an orchestration module can combine the ASR transcripts with speaker diarization outputs via the timing information from the two systems. Interestingly, we have found that this process can also be fully replaced by LLM. This can be achieved by explicitly including the timing information in the textual representation of both ASR transcripts and speaker diarization outputs. Additionally, more prompt engineering will be needed, such as explicitly explaining the format of the textual representation, and providing a one-shot example in the prompt. We show an example below.

Prompt:

Here we define the problem of speaker-transcript alignment. The transcript is represented by multiple entries of text, where each text has a starting time and an ending time. The speaker is also represented in this format. The alignment problem will assign a speaker to each word in the text, based on which speaker overlaps the most with that word in time.

Below is an example.

Transcript represented in format "[start - end] text":

[0 - 2.3] Good morning Patrick
 [2.5 - 5.2] how are you?
 [5.6 - 6.1] Good, good.
 [6.2 - 8.3] How are you Tom?
 [9.2 - 9.9] Pretty good.
 [10.0 - 11.1] Going to work?
 [12.5 - 13.6] Yes. Busy day.

Speaker represented in format "[start - end] speaker":

[0 - 5.1] <spk:1>
 [5.3 - 8.7] <spk:2>
 [9.2 - 10.9] <spk:1>
 [12.1 - 13.5] <spk:2>

The alignment result should be:

[0 - 2.3] <spk:1> Good morning Patrick
 [2.5 - 5.2] <spk:1> how are you?
 [5.6 - 6.1] <spk:2> Good, good.
 [6.2 - 8.3] <spk:2> How are you Tom?
 [9.2 - 9.9] <spk:1> Pretty good.
 [10.0 - 11.1] <spk:1> Going to work?
 [12.5 - 13.6] <spk:2> Yes. Busy day.

Based on the example above, we solve the alignment problem below.

Transcript represented in format "[start - end] text":

[0 - 5.6] Hi, how can I help you today?
 [6.2 - 11.1] Hi, I recently often feel quite dizzy at work.
 [11.6 - 15.5] Do you have any of these symptoms?
 [16.6 - 18.5] Like coughing, fever, or running nose?
 [20.0 - 21.1] No, I don't think so.
 [22.2 - 29.9] What do you usually eat for breakfast, lunch and dinner?
 [31.2 - 34.8] I'm on a diet, so not eating very much.
 [35.2 - 39.8] Mostly just one slice of toast in the morning, and eating salad for lunch and dinner.

Speaker represented in format "[start - end] speaker":

[0.3 - 5.3] <spk:1>
 [6.0 - 12.0] <spk:2>
 [12.9 - 20.1] <spk:1>
 [20.2 - 21.0] <spk:2>
 [21.8 - 31.1] <spk:1>
 [32.4 - 40.7] <spk:2>

The alignment result should be:

Completion:

Here's the aligned transcript:

[0 - 5.6] <spk:1> Hi, how can I help you today?
 [6.2 - 11.1] <spk:2> Hi, I recently often feel quite dizzy at work.
 [11.6 - 15.5] <spk:1> Do you have any of these symptoms?
 [16.6 - 18.5] <spk:1> Like coughing, fever, or running nose?
 [20.0 - 21.1] <spk:2> No, I don't think so.

[22.2 - 29.9] <spk:1> What do you usually eat for breakfast, lunch and dinner?
[31.2 - 34.8] <spk:2> I'm on a diet, so not eating very much.
[35.2 - 39.8] <spk:2> Mostly just one slice of toast in the morning, and eating salad for lunch and dinner.

To align the transcript and speaker segments, we assign each word in the transcript to the speaker whose segment overlaps the most with that word in time.